

Practical Implementation of the Kelly Criterion

Reproduction and Statistical Reassessment of Kelly Literature

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1 Introduction

The Kelly Criterion (Kelly 1956) prescribes the growth-optimal allocation $f^* = \Sigma^{-1}(\mu - r_f \mathbf{1})$ but is sensitive to estimation noise, the mean-vs-median divergence of long-horizon log-wealth, and deployment-side constraints (Breiman 1961; Thorp 2011). Carta and Conversano (2020) — hereafter CC20 — extend Kelly to multi-asset equity portfolios via a Taylor expansion of $\mathbb{E}[\log W]$ and report that aggressive Kelly multiples deliver superior out-of-sample (OOS) performance on EuroStoxx 50 versus tangent, MinVar, and equal-weight benchmarks. The claim is taken at face value: no confidence intervals, no hypothesis tests.

This study has four objectives. *(O1)* Validate Kelly geometry under known GBM parameters via Monte Carlo across four horizons. *(O2)* Replicate CC20’s pipeline — single-equity Banca Intesa, in-sample EuroStoxx 50 multi-asset Kelly, and the long-only OOS rolling-window backtest. *(O3)* Attach formal inference via HAC (Newey-West) tests on pairwise daily return differences. *(O4)* Probe robustness by lifting long-only constraints and repeating on Dow Jones Industrial Average (DJI).

The central finding is that CC20’s ranking is not robust. The long-only budget cap mechanically collapses Full, Double, and Triple Kelly onto an *identical* portfolio, so the apparent dominance of aggressive multiples is an optimizer artifact. HAC tests confirm it: only Half-vs-Full reaches significance ($p = 0.014$), with a 4 bp/day gap driven by the collapse. Lifting the cap separates the variants but degrades all of them (Sharpe 0.10–0.18, drawdowns 72–97%), tracking the volatility-pumping geometry the Monte Carlo predicts. On DJI, long-only Kelly is dominated by Buy-and-Hold on drawdown and all Kelly variants cluster tightly on Sharpe.

2 Data and Methodology

We use 35 surviving EuroStoxx 50 constituents with daily adjusted closes from Yahoo Finance (`tidyquant`, `quantmod`), January 2000 through December 2018, after a 20%-missing filter, forward-fill, and complete-case screening. Banca Intesa (ISP.MI) covers 2007–2018 daily; the DJI extension uses $\hat{\text{DJI}}$ over the same window. Monte Carlo paths are GBM with $\mu_{\text{ann}} = 0.12$, $\sigma_{\text{ann}} = 0.40$, $r_f = 0.01$ (theoretical full Kelly ≈ 0.6875); wealth evolves as $W_{t+1} = W_t[(1 - f)e^{r_f \Delta t} + f e^{r_t}]$.

The multi-asset Kelly QP follows CC20’s Taylor expansion of $\mathbb{E}[\log(1 + \mathbf{w}^\top \mathbf{r})]$:

$$\max_{\mathbf{w}} \mathbf{w}^\top (\mu - r_f \mathbf{1}) - \frac{1}{2} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = f, w_i \geq 0,$$

with $f \in \{0.5, 1, 2, 3\}$ for Half / Full / Double / Triple Kelly. Long-only clips f to $[0, 1]$; unconstrained allows shorts with leverage capped at $|f| \leq 3$. The QP is solved via `quadprog::solve.QP` after a `Matrix::nearPD` projection. Tangent and MinVar use the same machinery with $\mathbf{1}^\top \mathbf{w} = 1$. The OOS protocol is a 756-day rolling window with monthly re-optimization (`portfolioBacktest`, `lookback=252*3`, `optimize_every=rebalance_every=21`). For inference, daily return-differences $d_t = r_t^A - r_t^B$ are regressed on a constant with Newey-West HAC standard errors at five lags via `sandwich::NeweyWest` and `lmtest::coeftest`, testing $H_0 : \mathbb{E}[d_t] = 0$.

3 Results

3.1 Monte Carlo validation under known parameters

The four-horizon Monte Carlo isolates Kelly geometry from estimation noise. At 10,000 trades — roughly 40 years of daily trading — the divergence between mean and median terminal wealth across

multiples is severe (Table 1; sample wealth trajectories and full distributional shape in Appendix Figures A1 and A2). Triple Kelly’s mean of 518.5 sits on top of a 0.01 median and 79% probability of loss; Double Kelly posts a $184\times$ mean against a $1.67\times$ median; Full Kelly has the highest median (6.89) at 12% loss probability; only Half Kelly is unambiguous (4% loss probability, median 4.54).

Table 1: Monte Carlo terminal wealth at 10,000 trades (1,000 paths).

Strategy	Mean	Median	SD	P(Loss)	P(W>2)	P(W>10)
Half Kelly	7.03	4.54	9	4.0%	83.6%	20.0%
Full Kelly	31.47	6.89	94	12.3%	76.3%	41.7%
Double Kelly	183.80	1.67	1803	44.5%	48.3%	29.5%
Triple Kelly	518.50	0.01	11867	79.0%	17.5%	10.8%

Under log-normal wealth, $\mathbb{E}[\log W] < \log \mathbb{E}[W]$ by Jensen’s inequality and the gap widens with $f^2\sigma^2$. Aggressive multiples buy a larger mean by accepting a much smaller median: under known GBM parameters, Triple Kelly is not aggressive but actively destructive of typical investor outcomes. Any empirical case for Triple Kelly must clear a bar the noiseless simulation already shows it cannot.

3.2 Single-equity replication: Banca Intesa, 2007–2018

The Banca Intesa backtest (ISP.MI, daily, 2007–2018) reproduces CC20’s published single-asset ranking: Full Kelly delivers the highest terminal wealth (1.193, CAGR +1.53%) at -23.4% max drawdown; Half Kelly trails marginally (1.176, +1.40%) at -11.1% drawdown; Triple Kelly underperforms (-65.0% drawdown, terminal 0.937); and all three Kelly variants dominate Buy-and-Hold (0.619, -81.3%). Cumulative wealth paths are shown in Appendix Figure A3. The result is unsurprising — the static-fraction backtest amounts to a constant-rebalanced portfolio whose advantage over Buy-and-Hold under volatile single-asset paths is a textbook result (Cover 1991) — and the interest is whether multi-asset Kelly survives the OOS pipeline below.

3.3 In-sample EuroStoxx 50, 2000–2018

The in-sample multi-asset Kelly portfolio loads 62.3% on ITX.MC and 37.7% on VIV.PA — a two-asset corner solution typical of unconstrained Kelly under sample-mean inputs — and posts 19.07% annual return at 21.14% annual volatility, against tangent (16.41%/16.45%), MinVar (7.81%/12.25%), and equal-weight (7.80%/17.04%). The cumulative-wealth trajectory ends near $19\times$ initial capital against $\sim 12\times$ for tangent and $\sim 3\times$ for the benchmarks; the mean-variance plane with all four optimal portfolios is shown in Appendix Figure A4. The dominance is real but in-sample: the same sample mean and covariance the optimizer maximizes against drive the realized returns, so the comparison is not predictive — it is the OOS sections below that test whether it survives.

3.4 Out-of-sample EuroStoxx 50: long-only

Once we impose the long-only budget cap $\sum w_i = f \leq 1$, the constraint binds for any $f \geq 1$ and Full, Double, and Triple Kelly *collapse onto an identical portfolio* — same QP solution, same daily returns, same metrics. Table 2 makes the collapse visible: Full / Double / Triple Kelly produce identical Sharpe (0.621), max drawdown (0.639), and annual return (16.87%); only Half Kelly differs (Sharpe 0.556, return 7.87%, drawdown -35.2%); MinVar yields Sharpe 0.480.

Table 2: Long-only OOS performance, EuroStoxx 50, 2005-2018. Full / Double / Triple Kelly are identical: the budget constraint binds for $f \geq 1$.

Strategy	Sharpe	Max DD	Ann Return	Ann Vol	Sortino	CVaR (0.95)
Half Kelly	0.556	0.352	7.87%	14.15%	0.835	0.0195
Full Kelly	0.621	0.639	16.87%	27.17%	0.925	0.0380
Double Kelly	0.621	0.639	16.87%	27.17%	0.925	0.0380
Triple Kelly	0.621	0.639	16.87%	27.17%	0.925	0.0380
MinVar	0.480	0.580	8.55%	17.79%	0.685	0.0261

A naive reading endorses CC20: aggressive Kelly variants beat MinVar on Sharpe, return, and Sortino, with the collapsed-aggressive group reaching $\sim 6\times$ by 2018 against $\sim 2.5\times$ for Half Kelly and MinVar (Appendix Figure A5). Inference reverses the reading.

3.5 Statistical inference: HAC pairwise tests

Newey-West HAC tests (5 lags) on daily return differences $d_t = r_t^A - r_t^B$ deflate the apparent gaps. Only Half-vs-Full reaches conventional significance ($p = 0.014$), and even there the mean difference is just 4 bp/day. Half-vs-MinVar is essentially zero ($p = 0.86$); Full-vs-MinVar is +3 bp/day with $p = 0.14$. Pairwise mean differences with 95% HAC confidence intervals are visualized in Appendix Figure A6.

Table 3: HAC (Newey-West, 5 lags) tests on daily return differences, long-only OOS.

Pair	Mean Diff	HAC SE	t-stat	p-value
Half Kelly vs Full Kelly	-0.0004	0.0001	-2.4646	0.0138
Half Kelly vs MinVar	0.0000	0.0002	-0.1785	0.8584
Full Kelly vs MinVar	0.0003	0.0002	1.4736	0.1407

The Half-vs-Full significance is mechanically driven by the constraint collapse: Full Kelly’s “outperformance” is the difference between an $f = 0.5$ and an $f = 1$ corner solution under the same QP. Half-vs-MinVar and Full-vs-MinVar — the comparisons CC20’s narrative most directly relies on — both fail to reject. *Once realistic long-only constraints are imposed, the choice of Kelly multiplier has no statistically distinguishable effect on out-of-sample returns.* With $\sim 3,300$ daily observations across 13 years, this is the longest sample CC20’s data permits, and the failures to reject are therefore informative rather than data-starved. The absence of any such inferential check in CC20 is the structural weakness our extension targets.

3.6 Out-of-sample EuroStoxx 50: unconstrained

Lifting the budget cap separates the four Kelly variants but degrades all of them. Sharpe ratios fall to 0.10–0.18; max drawdowns climb to 72% (Half), 85% (Full), 96% (Double), 97% (Triple); annual returns are flat across multiples (5.5–7.5%) while volatility scales linearly with f from 31% to 62%. Appendix Figure A7 shows the Half Kelly allocation stack: period-long short positions in 2009–2010 and leverage hitting the $|f| = 3$ cap repeatedly. This is the over-leverage pathology the Monte Carlo predicts under known parameters: the long-only constraint is doing real work in the constrained backtest by forcing the optimizer away from Kelly geometry.

3.7 Generalization: Dow Jones Industrial Average

On the DJI single-index backtest (2010–2018), Buy-and-Hold posts a -20% max drawdown; Half and Full Kelly Long-Only also hold -20% ; the unconstrained Half through Triple Kelly variants run to -54% , -54% , -58% , and -65% . Sharpe ratios cluster tightly across all variants and Buy-and-Hold sits near the top of the range. The CC20 ranking does not generalize; the result that does is the conservative one — Half/Full Kelly Long-Only matches Buy-and-Hold on drawdown.

4 Discussion and Conclusion

Three findings emerge that CC20 do not report. (*F1*) The Monte Carlo establishes that Kelly’s mean-vs-median divergence is structural, not estimation-driven: aggressive multiples destroy median wealth even under known parameters. (*F2*) The long-only OOS budget cap mechanically collapses Full / Double / Triple Kelly onto an identical portfolio, so CC20’s headline ranking is evidence about a constraint, not about Kelly fractions. Lifting the cap separates the variants but degrades all of them in line with the Monte Carlo prediction — the consistency check confirming the pathology is geometry. (*F3*) HAC inference shows the only significant long-only pairwise gap is the constraint-collapse artifact; Full-vs-MinVar and Half-vs-MinVar — the comparisons CC20’s narrative most directly relies on — fail to reject on the longest sample CC20’s data permits.

Backtest comparisons are inferential exercises, and the absence of formal hypothesis tests in much of the practitioner literature lets cumulative-wealth panels do work they cannot honestly support. Half Kelly is the only fraction with a defensible record across all four diagnostics — Monte Carlo median wealth, single-equity drawdown, OOS long-only Sharpe, and DJI generalization — and the practical recommendation is the conservative one. Future work should attach formal inference to existing Kelly comparisons, explore μ/Σ shrinkage under rolling-window OOS, and test the constraint-collapse mechanism on additional equity panels.

5 References

- Breiman, L. (1961). Optimal gambling systems for favorable games. *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, 1, 65–78.
- Carta, A., & Conversano, C. (2020). Practical implementation of the Kelly Criterion. *Frontiers in Applied Mathematics and Statistics*, 6, 577050.
- Cover, T. M. (1991). Universal portfolios. *Mathematical Finance*, 1(1), 1–29.
- Kelly, J. L. (1956). A new interpretation of information rate. *Bell System Technical Journal*, 35(4), 917–926.
- MacLean, L. C., Thorp, E. O., & Ziemba, W. T. (Eds.). (2011). *The Kelly Capital Growth Investment Criterion*. World Scientific.
- Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Thorp, E. O. (2011). The Kelly criterion in blackjack, sports betting, and the stock market. In *The Kelly Capital Growth Investment Criterion* (pp. 789–832). World Scientific.